

SMOOTH COMPACTIFICATIONS WITH NONSEPARABLE REMAINDER UNDER $\mathfrak{b} = \mathfrak{c}$

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SOURCE AND TARGET

Drygier and Plebanek call a compactification $\gamma\omega$ *smooth* when the natural copy of c_0 is complemented in $C(\gamma\omega)$ [1]. For Boolean algebras $\text{Fin} \subseteq \mathcal{A} \subseteq P(\omega)$, their Lemma 3.1 says that $K_{\mathcal{A}}$ is smooth if and only if there is a bounded sequence of finitely additive measures (μ_n) on $\mathcal{A}$ such that

$$\mu_n|_{\text{Fin}} = 0, \quad \mu_n(A) - \delta_n(A) \longrightarrow 0 \quad (A \in \mathcal{A}).$$

The source proves under CH that a smooth compactification of ω can have nonseparable remainder. It also states that the authors do not know the ZFC answer and that their CH proof likely relaxes to $\mathfrak{b} = \mathfrak{c}$.

This packet proves that advertised relaxation.

THEOREM

Theorem 1. *Assume $\mathfrak{b} = \mathfrak{c}$. Then there is a smooth compactification $\gamma\omega$ of ω such that $\gamma\omega \setminus \omega$ is nonseparable.*

Equivalently, there is $\text{Fin} \subseteq \mathcal{A} \subseteq P(\omega)$ such that $\mathcal{A}/\text{Fin}$ is not σ -centered and $K_{\mathcal{A}}$ is smooth.

PROOF INTUITION

The source CH proof runs an ω_1 -stage induction through countable Boolean algebras. Countability has two roles: the key density-killing step can list old algebra elements, and CH lets the construction enumerate all countable dense subsets of the initial Cantor-like remainder.

Under $\mathfrak{b} = \mathfrak{c}$, $\mathfrak{c}$ is regular because $\mathfrak{b}$ is regular. Thus at every stage $\xi < \mathfrak{c}$, the current algebra has size $< \mathfrak{b}$. The countable listing in the source lemma can be replaced by the standard $\mathfrak{b}$ -domination finite-cover lemma used in [2].

There remains one bookkeeping trap: a countable dense set in the final compactum may have many possible extensions over intermediate algebras. The successor lemma below handles this by using two dense families. Sets chosen from the fixed initial dense family force every future extension of that initial family to contain the new set X . Sets chosen from a dense family in

the current remainder give the outer-measure estimate needed to extend the smoothness measures.

AUXILIARY LEMMAS

For $\text{Fin} \subseteq \mathcal{B} \subseteq P(\omega)$, write $K_{\mathcal{B}}$ for the Stone compactification and $K_{\mathcal{B}}^* = K_{\mathcal{B}} \setminus \omega$. If ν is a probability on $\mathcal{B}$ and $Z \subseteq \omega$, put

$$\nu_*(Z) = \sup\{\nu(B) : B \in \mathcal{B}, B \subseteq Z\}, \quad \nu^*(Z) = \inf\{\nu(B) : B \in \mathcal{B}, Z \subseteq B\}.$$

Lemma 2 (finite-cover lemma). *Let $|\mathcal{B}| < \mathfrak{b}$. For each $i \in \omega$, let $(S_i(k))_{k \in \omega}$ be a decreasing sequence in $\mathcal{B}$ with empty intersection. Then there is $g \in \omega^\omega$ such that, for*

$$X = \bigcup_{i \in \omega} S_i(g(i)),$$

every $B \in \mathcal{B}$ with $B \subseteq X$ is contained in a finite subunion $\bigcup_{i \leq N} S_i(g(i))$.

Proof. This is the finite-cover lemma from [2]; we recall the short proof. For finite m and $\varphi \in \omega^m$, write

$$X_\varphi = \bigcup_{i < m} S_i(\varphi(i)).$$

For fixed $B \in \mathcal{B}$, the set of $\varphi \in \omega^m$ such that $B \setminus X_\varphi$ is infinite has finitely many minimal elements under coordinatewise order, by induction on m . Hence there is a finite $I_m^B \subseteq \omega^m$ meeting $B \setminus X_\varphi$ whenever that set is infinite.

Choose $h_B \in \omega^\omega$ so that

$$S_i(h_B(i)) \cap \bigcup_{m \leq i} I_m^B = \emptyset \quad (i \in \omega).$$

Since $|\mathcal{B}| < \mathfrak{b}$, there is g eventually dominating all h_B . If $B \subseteq X$ and N is such that $h_B(i) \leq g(i)$ for $i \geq N$, then $B \setminus X_{g|N}$ cannot be infinite, because any point in $(B \setminus X_{g|N}) \cap I_N^B$ would have to lie in a tail $S_i(g(i))$, $i \geq N$, disjoint from I_N^B . Thus $B \setminus X_{g|N}$ is finite, and increasing N captures those finitely many points. $\square$

Lemma 3 (successor step). *Let $\text{Fin} \subseteq \mathcal{A}_0 \subseteq \mathcal{B} \subseteq P(\omega)$, with $\mathcal{A}_0$ countable and $|\mathcal{B}| < \mathfrak{b}$. Let $D = \{d_j : j \in \omega\}$ be a countable dense subset of $K_{\mathcal{A}_0}^*$, and let $Q = \{q_j : j \in \omega\}$ be a countable dense subset of $K_{\mathcal{B}}^*$. Suppose that (ν_n) is a sequence of nonatomic probabilities on $\mathcal{B}$ such that*

$$\nu_n|_{\text{Fin}} = 0, \quad \nu_n - \delta_n \rightarrow 0 \quad \text{pointwise on } \mathcal{B}.$$

Then there is an infinite $X \subseteq \omega$ such that:

- (i) $\omega \setminus X$ is infinite;
- (ii) every ultrafilter on $\mathcal{B}[X]$ whose restriction to $\mathcal{A}_0$ is one of the d_j 's contains X ;
- (iii) the ν_n 's extend to nonatomic probabilities $\tilde{\nu}_n$ on $\mathcal{B}[X]$ with $\tilde{\nu}_n - \delta_n \rightarrow 0$ pointwise on $\mathcal{B}[X]$.

Proof. Enumerate the disjoint union $D \sqcup Q$ as $(r_i)_{i \in \omega}$, retaining the information whether r_i is an ultrafilter on $\mathcal{A}_0$ or on $\mathcal{B}$. Choose positive numbers η_i with $\sum_i \eta_i < 1/2$.

For every i we build a decreasing sequence $S_i(k)$, $k \in \omega$, with empty intersection. If $r_i = d_j \in K_{\mathcal{A}_0}^*$, then

$$S_i(k) \in \mathcal{A}_0 \cap d_j.$$

If $r_i = q_j \in K_{\mathcal{B}}^*$, then

$$S_i(k) \in \mathcal{B} \cap q_j.$$

In both cases we also require

$$(1) \quad \nu_m(S_i(k)) < \eta_i \quad (m \leq k),$$

and

$$(2) \quad n \notin S_i(k) \implies \nu_n(S_i(k)) < \eta_i.$$

Assume $S_i(k-1)$ has been chosen, with $S_i(-1) = \omega$. Work in $\mathcal{A}_0$ if $r_i \in D$, and in $\mathcal{B}$ if $r_i \in Q$. Since the finite sum $\nu_0 + \dots + \nu_k$, restricted to that algebra, is nonatomic and r_i is nonprincipal, choose

$$C \in r_i, \quad C \subseteq S_i(k-1) \setminus \{0, \dots, k-1\},$$

with $\nu_m(C) < \eta_i$ for $m \leq k$. By pointwise convergence on C , the set

$$F = \{n \notin C : \nu_n(C) \geq \eta_i\}$$

is finite. The previous step implies $F \subseteq S_i(k-1)$, and the choice of C gives $F \cap \{0, \dots, k-1\} = \emptyset$. Put

$$S_i(k) = C \cup F.$$

Finite sets have ν_n -measure zero, so (1) and (2) hold. The sequence is decreasing and eventually avoids every integer.

Apply Lemma 2 to $(S_i(k))$, and set

$$X = \bigcup_i S_i(g(i)).$$

If $\omega \subseteq X$, the finite-cover property gives $\omega \subseteq \bigcup_{i \leq N} S_i(g(i))$ for some N , and hence

$$1 = \nu_0(\omega) \leq \sum_{i \leq N} \nu_0(S_i(g(i))) \leq \sum_{i \leq N} \eta_i < 1/2,$$

impossible. Thus $\omega \setminus X$ is infinite.

For every $d_j \in D$, one of the chosen sets $S_i(g(i))$ lies in d_j and is contained in X . Therefore every ultrafilter on $\mathcal{B}[X]$ extending d_j contains X . This proves (ii).

It remains to extend the measures. Fix n . If $B \in \mathcal{B}$ and $\nu_n(B) > 0$, then B is infinite. The clopen $\widehat{B} \cap K_{\mathcal{B}}^*$ is nonempty, so the dense set Q contains some q_j with $B \in q_j$. For the corresponding index i , both B and $S_i(g(i))$

lie in q_j , so $B \cap X \neq \emptyset$. Hence no positive- ν_n old algebra element is disjoint from X , and

$$(3) \quad \nu_n^*(X) = 1.$$

Now let $n \notin X$. If $B \in \mathcal{B}$ and $B \subseteq X$, then

$$B \subseteq \bigcup_{i \leq N} S_i(g(i))$$

for some N . For fixed $K \leq N$, split the right side into the first K pieces and the tail. By (2),

$$\nu_n \left(\bigcup_{K \leq i \leq N} S_i(g(i)) \right) \leq \sum_{i \geq K} \eta_i.$$

For each $i < K$, pointwise convergence on $S_i(g(i))$ gives $\nu_n(S_i(g(i))) \rightarrow 0$ along $n \notin X$. Thus

$$\limsup_{n \notin X} (\nu_n)_*(X) \leq \sum_{i \geq K} \eta_i,$$

and letting $K \rightarrow \infty$ gives

$$(4) \quad \lim_{n \notin X} (\nu_n)_*(X) = 0.$$

Define $t_n = 1$ for $n \in X$ and $t_n = (\nu_n)_*(X)$ for $n \notin X$. By (3), $t_n \in [(\nu_n)_*(X), \nu_n^*(X)]$. The Los–Marczewski extension theorem gives a probability extension $\tilde{\nu}_n$ to $\mathcal{B}[X]$ with $\tilde{\nu}_n(X) = t_n$. By (4), $\tilde{\nu}_n(X) - \delta_n(X) \rightarrow 0$, so the source convergence lemma [1, Lemma 3.3] yields

$$\tilde{\nu}_n - \delta_n \rightarrow 0 \quad \text{on } \mathcal{B}[X].$$

The extensions are nonatomic. If $u \in \text{ult}(\mathcal{B}[X])$ is principal, then a singleton in u has measure zero. Otherwise $u|_{\mathcal{B}}$ is nonprincipal, and nonatomicity of ν_n on $\mathcal{B}$ gives $B \in u|_{\mathcal{B}}$ with $\nu_n(B) < \varepsilon$. One of $B \cap X$ and $B \setminus X$ lies in u , and its $\tilde{\nu}_n$ -measure is at most $\nu_n(B)$. $\square$

PROOF OF THEOREM 1

Let $\mathcal{A}_0$ and (ν_n^0) be as in Lemma 6.2 of [1]: $\mathcal{A}_0$ is countable, $\text{Fin} \subseteq \mathcal{A}_0$, each ν_n^0 is a nonatomic probability vanishing on finite sets, and

$$\nu_n^0 - \delta_n \rightarrow 0 \quad \text{on } \mathcal{A}_0.$$

The space $K_{\mathcal{A}_0}^*$ is compact metrizable, so it has exactly $\mathfrak{c}$ many countable dense subsets. Enumerate them as

$$(D_\xi)_{\xi < \mathfrak{c}}.$$

We construct an increasing continuous chain $(\mathcal{A}_\xi, \nu_n^\xi)_{\xi < \mathfrak{c}}$. At limit stages take unions. Since $\mathfrak{b} = \mathfrak{c}$ and $\mathfrak{b}$ is regular, every $\mathcal{A}_\xi$ has cardinality $< \mathfrak{b}$.

At stage ξ , suppose $\mathcal{A}_\xi$ and (ν_n^ξ) have been constructed. If $K_{\mathcal{A}_\xi}^*$ is nonseparable, stop. Then $K_{\mathcal{A}_\xi}$ is already a smooth compactification by the measure criterion, and its remainder is nonseparable.

Otherwise choose a countable dense set $Q_\xi \subseteq K_{\mathcal{A}_\xi}^*$. Apply Lemma 3 with

$$\mathcal{B} = \mathcal{A}_\xi, \quad D = D_\xi, \quad Q = Q_\xi, \quad \nu_n = \nu_n^\xi.$$

Let X_ξ be the set produced by the lemma, put

$$\mathcal{A}_{\xi+1} = \mathcal{A}_\xi[X_\xi],$$

and let $\nu_n^{\xi+1}$ be the corresponding extensions. At a limit $\lambda < \mathfrak{c}$, set

$$\mathcal{A}_\lambda = \bigcup_{\xi < \lambda} \mathcal{A}_\xi,$$

and let ν_n^λ be the unique union of the coherent earlier measures. These limit measures are nonatomic because their restrictions to the fixed initial algebra $\mathcal{A}_0$ are nonatomic.

If the construction stops, we are done. Suppose it never stops, and define

$$\mathcal{A} = \bigcup_{\xi < \mathfrak{c}} \mathcal{A}_\xi, \quad \mu_n = \bigcup_{\xi < \mathfrak{c}} \nu_n^\xi.$$

For every $A \in \mathcal{A}$, the set A belongs to some stage, so

$$\mu_n(A) - \delta_n(A) \rightarrow 0.$$

Thus $K_{\mathcal{A}}$ is smooth.

We prove $K_{\mathcal{A}}^*$ is nonseparable. Suppose $E = \{e_j : j \in \omega\}$ were dense in $K_{\mathcal{A}}^*$. The restriction map $K_{\mathcal{A}}^* \rightarrow K_{\mathcal{A}_0}^*$ is onto, so

$$D = \{e_j|_{\mathcal{A}_0} : j \in \omega\}$$

is dense in $K_{\mathcal{A}_0}^*$. Hence $D = D_\xi$ for some $\xi < \mathfrak{c}$, after renumbering. At stage ξ , Lemma 3 made every future extension of a member of D_ξ contain X_ξ . Therefore every e_j contains X_ξ .

But $\omega \setminus X_\xi$ is infinite and belongs to $\mathcal{A}_{\xi+1}$, hence the clopen set

$$\widehat{\omega \setminus X_\xi} \cap K_{\mathcal{A}}^*$$

is nonempty. It is disjoint from E , contradicting the density of E . So $K_{\mathcal{A}}^*$ is nonseparable, and Theorem 1 follows.

NOVELTY AND LIMITATIONS

The run indexes had no previous full packet for arXiv:1601.03770. A bounded web search for the exact source title, “smooth compactification”, “nonseparable remainder”, and “ $\mathfrak{b} = \mathfrak{c}$ ” found the source paper [1] and the companion nonseparable-growth paper [2], but no later theorem proving the smooth $\mathfrak{b} = \mathfrak{c}$ strengthening.

This packet proves the $\mathfrak{b} = \mathfrak{c}$ version suggested in the source. It does not prove the stronger ZFC assertion.

HUMAN REVIEW RECOMMENDATION

First review Lemma 3. In particular, check that the $\mathcal{A}_0$ -pieces force all future extensions of D_ξ to contain X_ξ , while the current dense set Q_ξ is sufficient for $\nu_n^*(X_\xi) = 1$. Then review the final nonseparability contradiction.

REFERENCES

- [1] P. Drygier and G. Plebanek, *Compactifications of ω and the Banach space c_0* , arXiv:1601.03770, 2016.
- [2] P. Drygier and G. Plebanek, *Nonseparable growth of the integers supporting a measure*, arXiv:1501.06972, 2015.